

Configuración de Enrutador - R1 (Nivel 1)

Building configuration...

Current configuration : 1968 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

!

hostname R1

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

memory-size iomem 5

no ip icmp rate-limit unreachable

ip cef

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

archive

log config

hidekeys

!

!

ip tcp synwait-time 5

!

!

interface FastEthernet0/0

no ip address

duplex auto

speed auto

!

interface FastEthernet0/0.10

encapsulation dot1Q 10

ip address 192.168.50.49 255.255.255.240

!

interface FastEthernet0/0.20

encapsulation dot1Q 20

ip address 192.168.50.177 255.255.255.248

!

interface FastEthernet0/1

ip address 192.168.50.201 255.255.255.252

duplex auto

speed auto

!

```
interface FastEthernet1/0
```

```
no ip address
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface FastEthernet1/0.30
```

```
encapsulation dot1Q 30
```

```
ip address 192.168.50.185 255.255.255.248
```

```
!
```

```
interface FastEthernet1/0.40
```

```
encapsulation dot1Q 40
```

```
ip address 192.168.50.1 255.255.255.224
```

```
ip access-group 101 in
```

```
!
```

```
router ospf 1
```

```
log-adjacency-changes
```

```
passive-interface FastEthernet0/0.10
```

```
passive-interface FastEthernet0/0.20
```

```
passive-interface FastEthernet1/0.30
```

```
passive-interface FastEthernet1/0.40
```

```
network 192.168.50.0 0.0.0.31 area 0
```

```
network 192.168.50.48 0.0.0.15 area 0
```

```
network 192.168.50.176 0.0.0.7 area 0
```

```
network 192.168.50.184 0.0.0.7 area 0
```

```
network 192.168.50.200 0.0.0.3 area 0
```

```
!
```

```
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
access-list 101 remark BLOQUEO_WIFI_A_ZONAS_SEGURAS
access-list 101 deny ip any 192.168.50.192 0.0.0.7
access-list 101 deny ip any 192.168.50.32 0.0.0.15
access-list 101 permit ip any any
no cdp log mismatch duplex
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
```

!

!

End

Configuración de Enrutador - R2 (Nivel 2)

Building configuration...

Current configuration : 2105 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

!

hostname R2

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

memory-size iomem 5

no ip icmp rate-limit unreachable

ip cef

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

archive

log config

hidekeys

!

!

ip tcp synwait-time 5

!

!

interface FastEthernet0/0

ip address 192.168.50.202 255.255.255.252

duplex auto

speed auto

!

interface FastEthernet0/1

ip address 192.168.50.205 255.255.255.252

duplex auto

speed auto

!

interface FastEthernet1/0

no ip address

duplex auto

speed auto

!

```
interface FastEthernet1/0.50
  encapsulation dot1Q 50
  ip address 192.168.50.65 255.255.255.240
!
interface FastEthernet1/0.60
  encapsulation dot1Q 60
  ip address 192.168.50.161 255.255.255.248
!
interface FastEthernet2/0
  no ip address
  duplex auto
  speed auto
!
interface FastEthernet2/0.70
  encapsulation dot1Q 70
  ip address 192.168.50.169 255.255.255.248
!
interface FastEthernet2/0.80
  encapsulation dot1Q 80
  ip address 192.168.50.81 255.255.255.240
  ip access-group 101 in
!
router ospf 1
  log-adjacency-changes
  passive-interface FastEthernet1/0.50
  passive-interface FastEthernet1/0.60
```

```
passive-interface FastEthernet2/0.70
passive-interface FastEthernet2/0.80
network 192.168.50.64 0.0.0.15 area 0
network 192.168.50.80 0.0.0.15 area 0
network 192.168.50.160 0.0.0.7 area 0
network 192.168.50.168 0.0.0.7 area 0
network 192.168.50.200 0.0.0.3 area 0
network 192.168.50.204 0.0.0.3 area 0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
access-list 101 remark BLOQUEO_WIFI_A_ZONAS_SEGURAS
access-list 101 deny ip any 192.168.50.192 0.0.0.7
access-list 101 deny ip any 192.168.50.32 0.0.0.15
access-list 101 permit ip any any
no cdp log mismatch duplex
!
!
control-plane
!
!
line con 0
```



```
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
End
```

Configuración de Enrutador - R3 (Nivel 3)

Building configuration...

Current configuration : 2406 bytes

```
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R3
!
boot-start-marker
boot-end-marker
!
```

!

no aaa new-model

memory-size iomem 5

no ip icmp rate-limit unreachable

ip cef

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

archive

log config

hidekeys

!

!

ip tcp synwait-time 5

!

!

interface FastEthernet0/0

ip address 192.168.50.206 255.255.255.252

duplex auto

speed auto

!

interface FastEthernet0/1

no ip address

duplex auto

speed auto

!

interface FastEthernet0/1.140

encapsulation dot1Q 140

ip address 192.168.50.33 255.255.255.240

!

interface FastEthernet1/0

no ip address

duplex auto

speed auto

!

interface FastEthernet1/0.90

encapsulation dot1Q 90

ip address 192.168.50.81 255.255.255.240

!

interface FastEthernet1/0.120

encapsulation dot1Q 120

ip address 192.168.50.145 255.255.255.240

ip access-group 101 in

!

interface FastEthernet2/0

no ip address

duplex auto

speed auto

!

interface FastEthernet2/0.100

encapsulation dot1Q 100

ip address 192.168.50.129 255.255.255.240

!

interface FastEthernet2/0.110

encapsulation dot1Q 110

ip address 192.168.50.113 255.255.255.240

!

interface FastEthernet2/0.130

encapsulation dot1Q 130

ip address 192.168.50.193 255.255.255.248

!

router ospf 1

log-adjacency-changes

passive-interface FastEthernet0/1.140

passive-interface FastEthernet1/0.90

passive-interface FastEthernet1/0.120

passive-interface FastEthernet2/0.100

passive-interface FastEthernet2/0.110

passive-interface FastEthernet2/0.130

network 192.168.50.32 0.0.0.15 area 0

network 192.168.50.96 0.0.0.15 area 0

network 192.168.50.112 0.0.0.15 area 0

network 192.168.50.128 0.0.0.15 area 0

network 192.168.50.144 0.0.0.15 area 0

```
network 192.168.50.192 0.0.0.7 area 0
network 192.168.50.204 0.0.0.3 area 0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
access-list 101 remark BLOQUEO_WIFI_A_ZONAS_SEGURAS
access-list 101 deny ip any 192.168.50.192 0.0.0.7
access-list 101 deny ip any 192.168.50.32 0.0.0.15
access-list 101 permit ip any any
no cdp log mismatch duplex
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
```

logging synchronous

line vty 0 4

login

!

!

End

Tabla de Ruteo - Router 1

R1#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.168.50.0/24 is variably subnetted, 15 subnets, 4 masks

O 192.168.50.80/28

[110/11] via 192.168.50.202, 00:20:45, FastEthernet0/1

O 192.168.50.64/28

[110/11] via 192.168.50.202, 00:20:45, FastEthernet0/1

O 192.168.50.112/28

[110/21] via 192.168.50.202, 00:20:35, FastEthernet0/1

C 192.168.50.0/27 is directly connected, FastEthernet1/0.40

C 192.168.50.48/28 is directly connected, FastEthernet0/0.10

O 192.168.50.32/28

- [110/30] via 192.168.50.202, 00:20:37, FastEthernet0/1
- C 192.168.50.200/30 is directly connected, FastEthernet0/1
- O 192.168.50.204/30
- [110/20] via 192.168.50.202, 00:20:47, FastEthernet0/1
- O 192.168.50.192/29
- [110/21] via 192.168.50.202, 00:20:40, FastEthernet0/1
- O 192.168.50.144/28
- [110/21] via 192.168.50.202, 00:20:40, FastEthernet0/1
- O 192.168.50.128/28
- [110/21] via 192.168.50.202, 00:20:40, FastEthernet0/1
- C 192.168.50.184/29 is directly connected, FastEthernet1/0.30
- C 192.168.50.176/29 is directly connected, FastEthernet0/0.20
- O 192.168.50.168/29
- [110/11] via 192.168.50.202, 00:20:51, FastEthernet0/1
- O 192.168.50.160/29
- [110/11] via 192.168.50.202, 00:20:51, FastEthernet0/1

Tabla de Ruteo - Router 2

R2#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.168.50.0/24 is variably subnetted, 15 subnets, 4 masks

- C 192.168.50.80/28 is directly connected, FastEthernet2/0.80
- C 192.168.50.64/28 is directly connected, FastEthernet1/0.50
- O 192.168.50.112/28
[110/11] via 192.168.50.206, 00:20:59, FastEthernet0/1
- O 192.168.50.0/27 [110/11] via 192.168.50.201, 00:20:59, FastEthernet0/0
- O 192.168.50.48/28
[110/20] via 192.168.50.201, 00:20:59, FastEthernet0/0
- O 192.168.50.32/28
[110/20] via 192.168.50.206, 00:21:00, FastEthernet0/1
- C 192.168.50.200/30 is directly connected, FastEthernet0/0
- C 192.168.50.204/30 is directly connected, FastEthernet0/1
- O 192.168.50.192/29
[110/11] via 192.168.50.206, 00:21:00, FastEthernet0/1
- O 192.168.50.144/28
[110/11] via 192.168.50.206, 00:21:02, FastEthernet0/1
- O 192.168.50.128/28
[110/11] via 192.168.50.206, 00:21:02, FastEthernet0/1
- O 192.168.50.184/29
[110/11] via 192.168.50.201, 00:21:02, FastEthernet0/0
- O 192.168.50.176/29
[110/20] via 192.168.50.201, 00:21:03, FastEthernet0/0
- C 192.168.50.168/29 is directly connected, FastEthernet2/0.70
- C 192.168.50.160/29 is directly connected, FastEthernet1/0.60

Tabla de Ruteo - Router 3

R3#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.168.50.0/24 is variably subnetted, 16 subnets, 4 masks

O 192.168.50.80/28

[110/11] via 192.168.50.205, 00:00:13, FastEthernet0/0

O 192.168.50.64/28

[110/11] via 192.168.50.205, 00:26:05, FastEthernet0/0

C 192.168.50.112/28 is directly connected, FastEthernet2/0.110

C 192.168.50.96/28 is directly connected, FastEthernet1/0.90

O 192.168.50.0/27 [110/21] via 192.168.50.205, 00:26:05, FastEthernet0/0

O 192.168.50.48/28

[110/30] via 192.168.50.205, 00:26:07, FastEthernet0/0

C 192.168.50.32/28 is directly connected, FastEthernet0/1.140

O 192.168.50.200/30

[110/20] via 192.168.50.205, 00:26:07, FastEthernet0/0

C 192.168.50.204/30 is directly connected, FastEthernet0/0

C 192.168.50.192/29 is directly connected, FastEthernet2/0.130

C 192.168.50.144/28 is directly connected, FastEthernet1/0.120

C 192.168.50.128/28 is directly connected, FastEthernet2/0.100

O 192.168.50.184/29

[110/21] via 192.168.50.205, 00:26:09, FastEthernet0/0

O 192.168.50.176/29

[110/30] via 192.168.50.205, 00:26:10, FastEthernet0/0

O 192.168.50.168/29

[110/11] via 192.168.50.205, 00:26:11, FastEthernet0/0

O 192.168.50.160/29

[110/11] via 192.168.50.205, 00:26:11, FastEthernet0/0

Pruebas de Ping

PCV1>

PCV1> ping 192.168.50.194

192.168.50.194 icmp_seq=1 timeout

84 bytes from 192.168.50.194 icmp_seq=2 ttl=61 time=92.150 ms

84 bytes from 192.168.50.194 icmp_seq=3 ttl=61 time=91.693 ms

84 bytes from 192.168.50.194 icmp_seq=4 ttl=61 time=93.542 ms

84 bytes from 192.168.50.194 icmp_seq=5 ttl=61 time=91.497 ms

PCV1> ping 192.168.50.114

192.168.50.114 icmp_seq=1 timeout

192.168.50.114 icmp_seq=2 timeout

84 bytes from 192.168.50.114 icmp_seq=3 ttl=61 time=92.454 ms

84 bytes from 192.168.50.114 icmp_seq=4 ttl=61 time=91.906 ms

84 bytes from 192.168.50.114 icmp_seq=5 ttl=61 time=92.190 ms

PCV1> ping 192.168.50.83

192.168.50.83 icmp_seq=1 timeout

192.168.50.83 icmp_seq=2 timeout

84 bytes from 192.168.50.83 icmp_seq=3 ttl=62 time=60.620 ms

84 bytes from 192.168.50.83 icmp_seq=4 ttl=62 time=61.048 ms

84 bytes from 192.168.50.83 icmp_seq=5 ttl=62 time=61.725 ms

Prueba de Trace

PCV1> trace 192.168.50.194

trace to 192.168.50.194, 8 hops max, press Ctrl+C to stop

1 192.168.50.49 15.749 ms 16.428 ms 16.887 ms

2 192.168.50.202 46.545 ms 46.744 ms 46.157 ms

3 192.168.50.206 77.018 ms 76.415 ms 77.790 ms

4 * * *

5 *192.168.50.194 92.795 ms (ICMP type:3, code:3, Destination port unreachable)

Prueba ACLs

VPCS> ping 192.168.50.194

*192.168.50.1 icmp_seq=1 ttl=255 time=16.094 ms (ICMP type:3, code:13, Communication administratively prohibited)

*192.168.50.1 icmp_seq=2 ttl=255 time=15.893 ms (ICMP type:3, code:13, Communication administratively prohibited)

*192.168.50.1 icmp_seq=3 ttl=255 time=16.224 ms (ICMP type:3, code:13, Communication administratively prohibited)

*192.168.50.1 icmp_seq=4 ttl=255 time=15.776 ms (ICMP type:3, code:13, Communication administratively prohibited)

*192.168.50.1 icmp_seq=5 ttl=255 time=15.936 ms (ICMP type:3, code:13, Communication administratively prohibited)